Politechnika
Śląska

Starbucks Customer Ordering Patterns

Analysis of the "Digital Venti Effect"

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Chapter 1

General description of the dataset

The subject of the project is a statistical analysis of the "Starbucks Customer Ordering Patterns" dataset. This dataset, which was published on Kaggle, consists of 100 000 real transactions made by Starbucks customers over a two-year period (2024-2025). It allows to analyse customer behavior, focusing on how orders are placed, the financial value of those orders, and the time required for fulfillment. To ensure the methods are adjusted to the dataset, five key features have been isolated to analyze trends in spending, demographic preferences, and operational efficiency.

1.1 All attributes

- IDs:
  - customer _id: unique identifier for each customer (15000 unique profiles),
  - order_id: unique transaction id (100 000 orders),
  - store_id: identifier for each store,
- Date:
  - order_date: (ranging from 01.2024 to 12.2025),
  - order_time,
  - day_of_week,
- Order:
  - order_channel: method used to place the order (Mobile App, Drive-Thru, In-Store Cashier or Kiosk),
  - store_location_type: (Urban, Suburban or Rural)
  - region: (Northeast, Southeast, Midwest, West or Southwest),
  - customer_age_group: (18-24, 25-34, 35-44, 45-54 or 55+),
  - customer_gender: (Male, Female or Other),

- is_rewards_member: whether the customer is a member of the Starbucks Rewards program (True or False),
- cart_size: number of items in the order,
- num_customizations: number of modifications made to the order (extra shots, syrup, different milk),
- total_spend: total cost of the transaction in USD,
- fulfillment_time_min: time (in minutes) from the moment the order is placed until it is ready for the customer,
- drink_category: (Coffee, Tea, Refresher or Other),
- has_food_item: (True or False),
- order_ahead: whether the order was placed ahead of time (True or False),
- customer_satifsaction: rating from 1 to 5 (correlated with fulfillment time and channel),

1.2 Summary of selected attributes

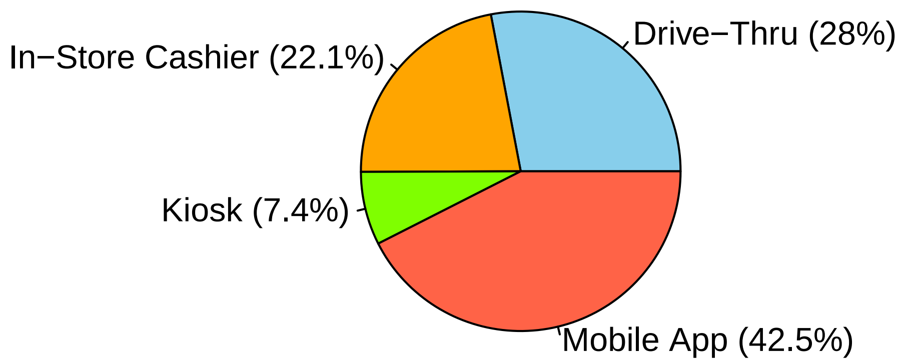
1.2.1 order_channel

description

This feature lists which method of ordering was used by the customer while placing order. Available options were Mobile App, Drive-Thru, In-Store Cashier, Kiosk.

graph representation

Figure 1.1: Percentage distribution of order channels



use-case

We will use this feature to check how customers were placing orders.

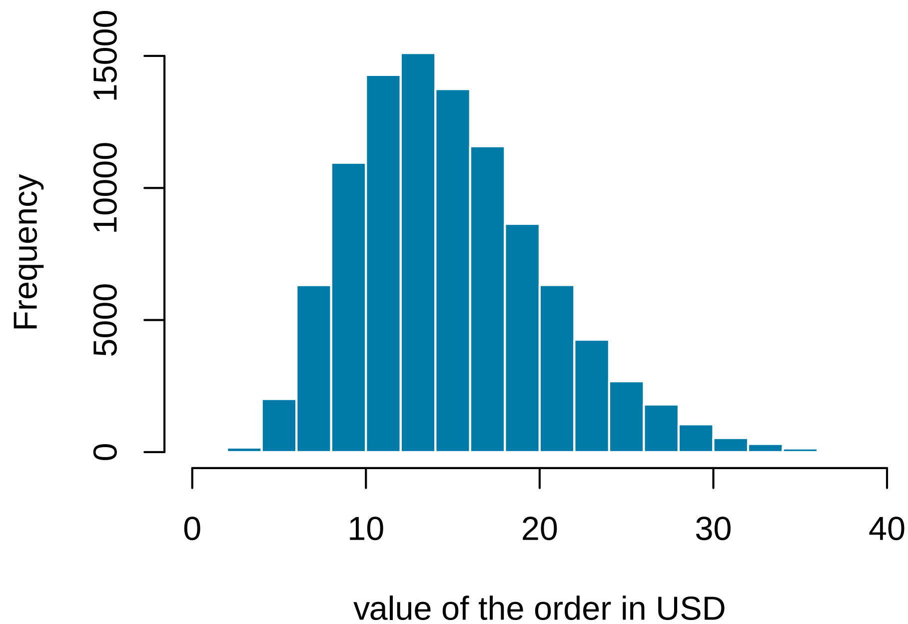
1.2.2 total_spend

description

This feature shows the total cost of the transaction in USD.

graph representation

Figure 1.2: Distribution of total spend among customers



use-case

We will use this data and compare it to the order_channel to check if there exists a relation between the method of ordering and the amount spent.

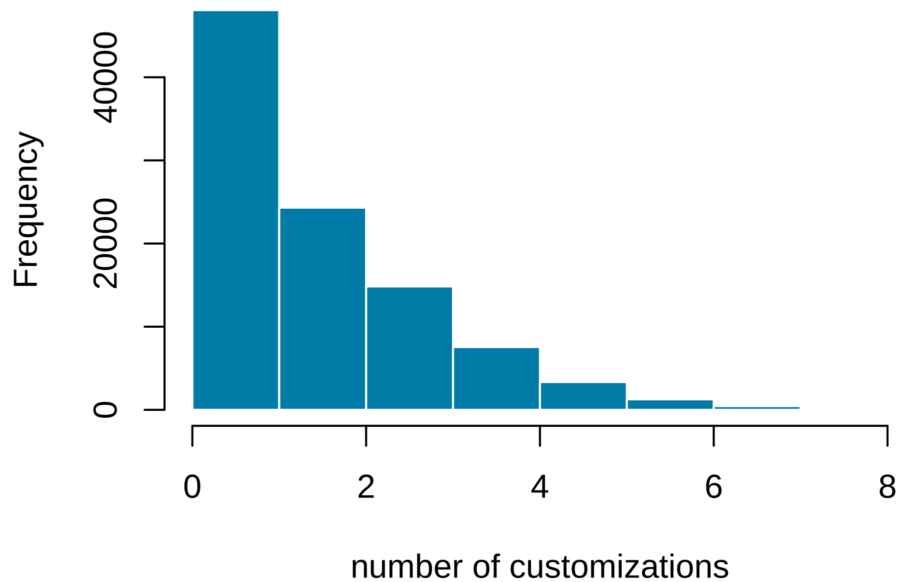
1.2.3 num_customizations

description

This feature shows the number of modifications made to the order (e.g., extra shots, syrup, milk sub).

graph representation

Figure 1.3: Distribution of the number of customizations among customers



use-case

We will use this data to check if mobile app users demonstrate higher tendency to customizing their orders.

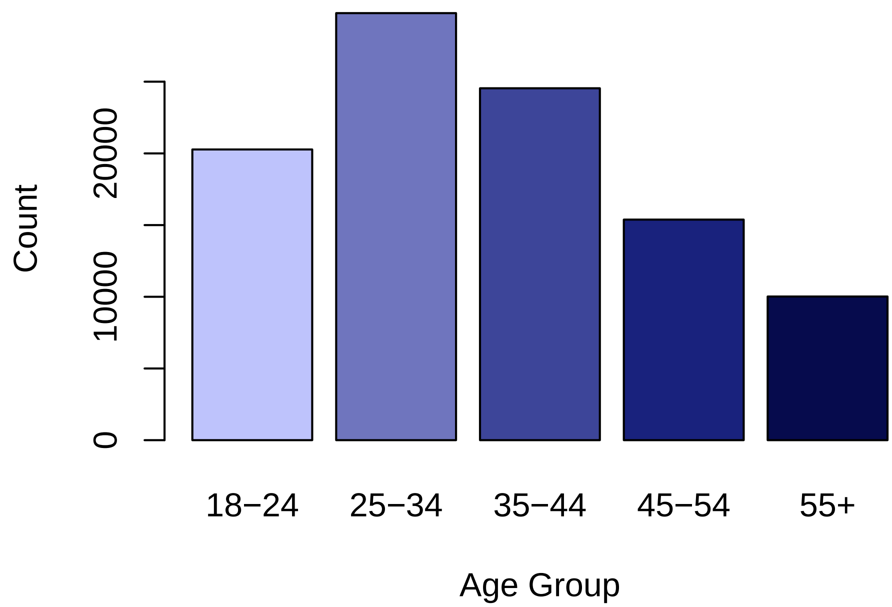
1.2.4 customer_age_group

description

This feature shows the age group of the customer.

graph representation

Figure 1.4: Distribution of customer age groups



use-case

A relationship between customer age groups and their ordering behavior will be investigated, notably whether certain age groups prefer eg. specific order channels or spend more money.

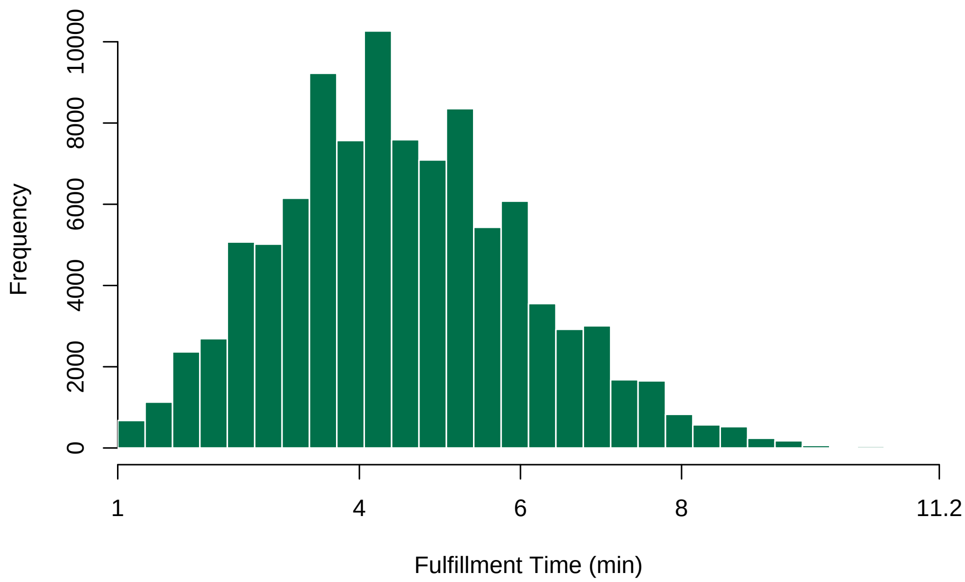
1.2.5 fulfillment_time_min

description

This feature measures the time (in minutes) from the moment the order is placed until it is ready for the customer. It is a key metric for operational efficiency.

graph representation

Figure 1.5: Distribution of fulfillment time among customers



use-case

We will analyze this attribute to determine how the complexity of an order (number of customizations) affects service speed. This helps identify whether encouraging modifications leads to operational bottlenecks.

Chapter 2

Calculation of descriptive statistics

2.1 Descriptive statistics by attribute

2.1.1 order_channel

As the order_channel is presented as a string, we will calculate the mode and cardinality to understand the distribution of ordering methods among customers. This will help us identify which channels are most popular and if there are any significant differences in customer preferences.

Table 2.1: Descriptive statistics for order_channel

Measure of a central tendency	Obtained result
mode	25-34
cardinality	5

2.1.2 total_spend

As the total_spend is a continuous numerical variable, we will calculate the mean, median, mode, standard deviation, range, 0.25 quartile, and 0.75 quartile to understand the distribution of spending amounts among customers. This will help us identify average spending patterns, variability in transaction values, and whether there are significant outliers in customer expenditures.

Table 2.2: Descriptive statistics for total_spend

Measure of a central tendency	Obtained result
mean value	14.8667708
median value	14.17
mode value	12.1453342509566
range	3.51 - 40.31
standard deviation	5.50680009558536
0.25 Quartile	10.8375
0.75 Quartile	18.18

2.1.3 num_customizations

As num_customizations is a discrete numerical variable, we will calculate the mean, median, mode, standard deviation, range, 0.25 quartile, and 0.75 quartile to understand the distribution of order modifications among customers. This will help us identify average customization patterns, variability in the number of modifications per order, and whether certain customers tend to customize their orders significantly more than others.

Table 2.3: Descriptive statistics for num_customizations

Measure of a central tendency	Obtained result
mean value	1.81077
median value	2
mode value	0.998356888479198
range	0 - 8
standard deviation	1.46279985129078
0.25 Quartile	1
0.75 Quartile	3

2.1.4 customer_age_group

As the customer_age_group is presented as a string, we will calculate the mode and cardinality to understand the distribution of age groups among customers. This will help us identify which age groups are most represented and if there are any significant differences in customer preferences.

Table 2.4: Descriptive statistics for customer_age_group

Measure of a central tendency	Obtained result
mode	25-34
cardinality	5

2.1.5 fulfillment_time_min

As the fulfillment_time_min is a continuous numerical variable, we will calculate the mean, median, mode, standard deviation, range, 0.25 quartile, and 0.75 quartile to understand the distribution of fulfillment times among customers. This will help us identify average fulfillment patterns, variability in delivery times, and whether there are significant outliers in customer delivery experiences.

Table 2.5: Descriptive statistics for fulfillment_time_min

Measure of a central tendency	Obtained result
mean value	4.54608
median value	4.4
mode value	4.31807337490211
range	1 - 11.2
standard deviation	1.55026948204314
0.25 Quartile	3.4
0.75 Quartile	5.5

Chapter 3

Attributes visualization

Chapter 4

Testing of the hypothesis about the normality of distributions

Chapter 5

Expected relationships between the attributes

Chapter 6

Additional analyses aka something new

Chapter 7

Results and conclusions